

Application Build and Release Automation Lab

Assignment / Practice 3 — Library Management System

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Subject: Application Build and Release Automation

Semester: 5th

Submitted Evidence: Q1–Q5 terminal screenshots

Q1 — Linux Workspace

```
abhijectsingh@Abhijects-MacBook-Air Abhijeet-GLA-HealthSystem % cd ~/Cloud-Assignment
abhijectsingh@Abhijects-MacBook-Air Cloud-Assignment % mkdir Abhijeet-GLA-LibrarySystem
abhijectsingh@Abhijects-MacBook-Air Cloud-Assignment % cd Abhijeet-GLA-LibrarySystem
abhijectsingh@Abhijects-MacBook-Air Abhijeet-GLA-LibrarySystem % mkdir src docs backup
abhijectsingh@Abhijects-MacBook-Air Abhijeet-GLA-LibrarySystem % touch library_info.txt
abhijectsingh@Abhijects-MacBook-Air Abhijeet-GLA-LibrarySystem % nano library_info.txt
abhijectsingh@Abhijects-MacBook-Air Abhijeet-GLA-LibrarySystem % pwd
ls
cat library_info.txt
tree
/Users/abhijectsingh/Cloud-Assignment/Abhijeet-GLA-LibrarySystem
backup      docs      library_info.txt      src
Student Name: Abhijeet Singh
Roll Number: 2415001871
Subject Name: Application Build and Release Automation
Project Name: Library Management System
Semester: 5th
.
├── backup
├── docs
├── library_info.txt
└── src

4 directories, 1 file
abhijectsingh@Abhijects-MacBook-Air Abhijeet-GLA-LibrarySystem % █
```

Q2 — Git Repository

```
4 directories, 1 file
abhiijeetsingh@Abhijects-MacBook-Air Abhijeet-GLA-LibrarySystem % git init
Initialized empty Git repository in /Users/abhiijeetsingh/Cloud-Assignment/Abhijeet-GLA-LibrarySystem/.git/
abhiijeetsingh@Abhijects-MacBook-Air Abhijeet-GLA-LibrarySystem %
abhiijeetsingh@Abhijects-MacBook-Air Abhijeet-GLA-LibrarySystem % git config user.name "Abhiijeet Singh"
git config user.email "singhabhiijeet9454@gmail.com"
abhiijeetsingh@Abhijects-MacBook-Air Abhijeet-GLA-LibrarySystem %
abhiijeetsingh@Abhijects-MacBook-Air Abhijeet-GLA-LibrarySystem % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    library_info.txt

nothing added to commit but untracked files present (use "git add" to track)
abhiijeetsingh@Abhijects-MacBook-Air Abhijeet-GLA-LibrarySystem %
abhiijeetsingh@Abhijects-MacBook-Air Abhijeet-GLA-LibrarySystem % git add .
abhiijeetsingh@Abhijects-MacBook-Air Abhijeet-GLA-LibrarySystem % git commit -m "Initial project setup"
[main (root-commit) e8624d1] Initial project setup
 1 file changed, 5 insertions(+)
 create mode 100644 library_info.txt
abhiijeetsingh@Abhijects-MacBook-Air Abhijeet-GLA-LibrarySystem % █
```

Q3 — Feature Branch

```
1 file changed, 4 insertions(+)
labhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git log --oneline
3d54488 (HEAD -> main) Update library information
e8624d1 Initial project setup
labhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git checkout -b book-feature
Switched to a new branch 'book-feature'
labhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % touch src/book.txt
nano src/book.txt
labhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git add src/book.txt
git commit -m "Add book management module"
[book-feature e38ef2f] Add book management module
1 file changed, 11 insertions(+)
create mode 100644 src/book.txt
labhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git branch
* book-feature
main
labhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git branch --show-current
book-feature
labhijeetsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % █
```



Q3 — Return to Main / Feature Independence

```
* Learn something & share it with the world
abhiijeetsingh@Abhiijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git log --oneline
3d54488 (HEAD -> main) Update library information
e8624d1 Initial project setup
abhiijeetsingh@Abhiijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git checkout -b book-feature
Switched to a new branch 'book-feature'
abhiijeetsingh@Abhiijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % touch src/book.txt
[nano src/book.txt]
abhiijeetsingh@Abhiijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git add src/book.txt
git commit -m "Add book management module"
[book-feature e38ef2f] Add book management module
1 file changed, 11 insertions(+)
create mode 100644 src/book.txt
abhiijeetsingh@Abhiijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git branch
* book-feature
main
abhiijeetsingh@Abhiijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git branch --show-current
book-feature
abhiijeetsingh@Abhiijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git checkout main
Switched to branch 'main'
abhiijeetsingh@Abhiijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git branch --show-current
main
abhiijeetsingh@Abhiijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % ls src
ls: src: No such file or directory
abhiijeetsingh@Abhiijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % █
```

Q4 — Merge Conflict Generated

```
* 88924d1 Initial project setup
abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git checkout -b library-update
Switched to a new branch 'library-update'
abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % nano library_info.txt
abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git add library_info.txt
git commit -m "Update project name in library branch"
[library-update 4d85286] Update project name in library branch
1 file changed, 1 insertion(+), 1 deletion(-)
abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git checkout main
Switched to branch 'main'
abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % nano library_info.txt
abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git add library_info.txt
git commit -m "Update project name in main branch"
[main 7a6cad7] Update project name in main branch
1 file changed, 1 insertion(+), 1 deletion(-)
abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git merge library-update
Auto-merging library_info.txt
CONFLICT (content): Merge conflict in library_info.txt
Automatic merge failed; fix conflicts and then commit the result.
abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git status
On branch main
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
        both modified:   library_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % cat library_info.txt
Student Name: Abhijeet Singh
Roll Number: 2415001571
Subject Name: Application Build and Release Automation
<<<<<< HEAD
Project Name: Smart Library Management System
=====
Project Name: Advanced Library Management System
>>>>>> library-update
Semester: 5th
Project Objective: To manage library books and provide an efficient system for book management.
Project Description: A Git-based library management system for maintaining and managing book information.
abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % █
```

Q4 — Conflict Resolved & History

```
Switched to branch 'main'
abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % nano library_info.txt
abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git add library_info.txt
git commit -m "Update project name in main branch"
[main 7a6cad7] Update project name in main branch
1 file changed, 1 insertion(+), 1 deletion(-)
abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git merge library-update
Auto-merging library_info.txt
CONFLICT (content): Merge conflict in library_info.txt
Automatic merge failed; fix conflicts and then commit the result.
abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git status
On branch main
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
        both modified:   library_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % cat library_info.txt
Student Name: Abhijeet Singh
Roll Number: 2415001871
Subject Name: Application Build and Release Automation
<----- HEAD
Project Name: Smart Library Management System
=====
Project Name: Advanced Library Management System
>>>>>> library-update
Semester: 5th
Project Objective: To manage library books and provide an efficient system for book management.
Project Description: A Git-based library management system for maintaining and managing book information.
abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % nano library_info.txt
abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git add library_info.txt
abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git status
On branch main
All conflicts fixed but you are still merging.
  (use "git commit" to conclude merge)

Changes to be committed:
        modified:   library_info.txt

abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git commit -m "Resolve library information merge conflict"
[main 37c3751] Resolve library information merge conflict
abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % cat library_info.txt
Student Name: Abhijeet Singh
Roll Number: 2415001871
Subject Name: Application Build and Release Automation
Project Name: Smart Advanced Library Management System
Semester: 5th
Project Objective: To manage library books and provide an efficient system for book management.
Project Description: A Git-based library management system for maintaining and managing book information.
abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem %
abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % cat library_info.txt
Student Name: Abhijeet Singh
Roll Number: 2415001871
Subject Name: Application Build and Release Automation
Project Name: Smart Advanced Library Management System
Semester: 5th
Project Objective: To manage library books and provide an efficient system for book management.
Project Description: A Git-based library management system for maintaining and managing book information.
abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git status
On branch main
nothing to commit, working tree clean
abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % git log --oneline --all --graph
* 37c3751 (HEAD -> main) Resolve library information merge conflict
|
| * 4d85206 (library-update) Update project name in library branch
| * | 7a6cad7 Update project name in main branch
|/
* e38ef2f (book-feature) Add book management module
* 3d54480 Update library information
* e9624d1 Initial project setup
abhijectsingh@Abhijeets-MacBook-Air Abhijeet-GLA-LibrarySystem % █
```

Q5 — Final Demonstration

```
abhijeetsingh@Abhijects-MacBook-Air Abhijeet-GLA-LibrarySystem % pwd
/Users/abhijeetsingh/Cloud-Assignment/Abhijeet-GLA-LibrarySystem
abhijeetsingh@Abhijects-MacBook-Air Abhijeet-GLA-LibrarySystem % tree
.
├── backup
├── docs
├── library_info.txt
└── src
    └── book.txt

4 directories, 2 files
abhijeetsingh@Abhijects-MacBook-Air Abhijeet-GLA-LibrarySystem % git status
On branch main
nothing to commit, working tree clean
abhijeetsingh@Abhijects-MacBook-Air Abhijeet-GLA-LibrarySystem % git branch
book-feature
library-update
* main
abhijeetsingh@Abhijects-MacBook-Air Abhijeet-GLA-LibrarySystem % git log --oneline --all --graph
* 57c3751 (HEAD -> main) Resolve library information merge conflict
|
| * 4d85286 (library-update) Update project name in library branch
| * | 7a6cad7 Update project name in main branch
|/
* e38ef2f (book-feature) Add book management module
* 3d54488 Update library information
* e8a2ad5 Initial project setup
abhijeetsingh@Abhijects-MacBook-Air Abhijeet-GLA-LibrarySystem % cat library_info.txt
Student Name: Abhijeet Singh
Roll Number: 2415001871
Subject Name: Application Build and Release Automation
Project Name: Smart Advanced Library Management System
Semester: 8th
Project Objective: To manage library books and provide an efficient system for book management.
Project Description: A Git-based library management system for maintaining and managing book information.
abhijeetsingh@Abhijects-MacBook-Air Abhijeet-GLA-LibrarySystem % █
```